Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian Paleomagnetic Field for North America

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The natural remanent magnetizations of Late Cambrian siltstones of the Wilberns Formation have been analyzed with alternating field, thermal, and chemical demagnetization techniques. From mainly thermal demagnetization analyses a mean direction of magnetization has been calculated: declination D of 101.5°, inclination I of +13.5°, k = 171, and α_{95} = 4.6° for seven sites (55 samples), yielding a virtual pole position at 6.4°N, 158.8°E, dp = 2.4°, and dm = 4.7°. This polar error (dp, dm) represents the oval of 95% confidence about the pole position. Comparisons with previously published Late Cambrian data reveal good agreement between the thermally isolated directions of magnetization. The difference in the published results obtained through alternating field demagnetization can be explained by a late Paleozoic remagnetization or by a previously unexpected Cambrian shift or loop of the apparent polar wander path.

INTRODUCTION

This paper describes paleomagnetic measurements and demagnetization analyses of the Late Cambrian Point Peak Member of the Wilberns Formation of the Llano uplift area in Texas and compares the results with previous work. A survey of published early Paleozoic paleomagnetic pole positions for North America reveals their relative scarcity [*McElhinny*, 1973]. Thus a comparison between North American data and results from other continental blocks is hardly possible without new data, and theories for pre-Mesozoic relative drift between the major continents [*Wilson*, 1966; *Dewey and Bird*, 1970; *McKerrow and Ziegler*, 1972; and many others] lack adequate paleomagnetic evidence.

Despite various efforts to obtain unambiguous Cambrian paleomagnetic pole positions for the North American craton, the few results that have been published are scattered and inconclusive. Several reasons can be cited for this. Early results [*Day and Runcorn*, 1955; *Howell and Martinez*, 1957; *Collinson and Runcorn*, 1960] were not obtained through detailed demagnetization analysis; consequently, such results may be contaminated by secondary magnetizations and have sometimes been shown to be in error by more than 40°. Generally, these data are now discarded, and we will not discuss them here. Another reason for the scattering of the Cambrian poles may be apparent polar wander during the 100 m.y. of this period; it is conceivable that North America moved considerably during this time with respect to the pole. Yet another reason may be found in the demagnetization analysis of the paleomagnetic samples when it is not recognized that more than one vectorial component of magnetization is present. Natural remanent magnetizations (NRM's) of older rocks are often of this multivectorial nature. In a later section we will compare our data with other Cambrian poles published for North America and discuss these problems further.

Finally, we present a list of preferred pole positions for the early Paleozoic and discuss the paleogeographic position of North America during the Cambrian.

GEOLOGY AND SAMPLING

The Llano uplift area in central Texas (see map of Figure 1) consists of a dome-shaped upwarping of Precambrian basement and Paleozoic sediments, roughly 120 km in diameter, which was initiated in Mississippian time. Paleozoic rocks thus are distributed in a roughly circular pattern peripheral to a central Precambrian igneous and metamorphic complex and represent deposition from Late Cambrian to Early Pennsylvanian times. Structural deformation was primarily represented by gentle tilting of the strata radially outward from the uplift. Moreover, numerous faults exist, most of which are normal [*Bell and Barnes*, 1972].

The Wilberns Formation is subdivided into four members from base to top: the Welge Sandstone Member, the Morgan Creek Limestone Member, the Point Peak Member, and the San Saba Member. Paleontological evidence has dated the Point Peak as Franconian to Early Trempealeauan (Middle Croixian), and the Cambrian-Ordovician boundary has been placed within the San Saba [*Bell and Barnes*, 1972]. The Point Peak Member ranges from 8 m in thickness in the southeast area of the uplift to 72 m in the northwest. It consists of a variegated siltstone in the lower part and is highly calcareous in the upper part, where it ranges from silty limestone to large stromatolitic bioherms, which coalesce into biostromes in places. Numerous shale laminae are present throughout the Point Peak, and glauconite is a common constituent among the siltstones; other Fe minerals identified are goethite and/or lepidocrocite and hematite. The bedding is thin, from 1 to 15 cm, averaging approximately 6 cm; bioturbation has been common in the more calcareous units; and ripple marks are numerous within the siltstones, frequently making the distinct bedding surfaces slightly irregular.

A total of 63 samples distributed among eight sites was collected from within the brown, yellow, and red Point Peak siltstone units. The locations of these sites (Figure 1) are restricted to the northwest region of the uplift for two reasons. First, the Point Peak as a whole is poorly exposed; good exposures were found primarily in the northwest. Second, of those exposures found on the southwest and southeast perimeters of the uplift, none were located with siltstones of sufficient competency to sample. Because of the thin bedding and fragility of most outcrops, hand samples were required. Special care was taken to avoid slumped outcrops, and bioturbated units were also avoided. The strata sampled were nearly horizontal with dips generally less than 10°, except for site 7, where dips ranged from 14° to 23°. Numerous northeast-trending normal

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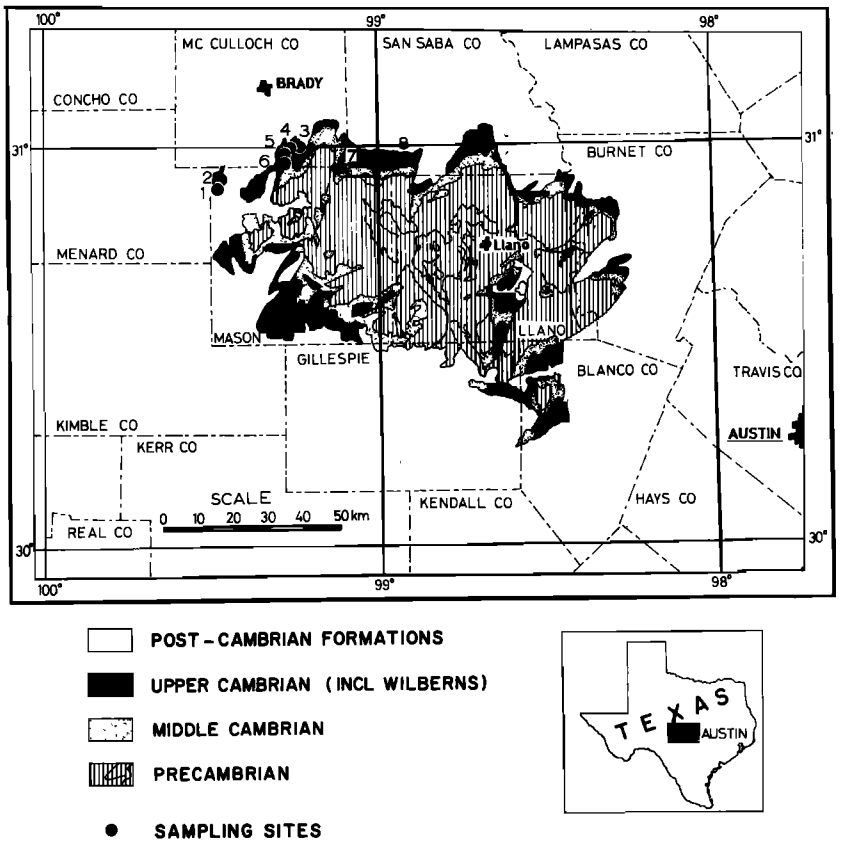


Fig. 1. Schematic geologic map of the Llano uplift, Texas. The eight sampling sites of the Point Peak Member of the Wilberns Formation are shown.

faults tilt the sampled strata away from a radially outward attitude, but no folding was observed in the sampling area.

Measurement and Demagnetization

Measurement of the NRM was carried out with a Schonstedt spinner magnetometer and a ScT cryogenic magnetometer. Samples from seven of the eight sites ranged in intensity from 34×10^{-6} emu/cm^3 to 3×10^{-6} emu/cm^3. Site 3 had intensities about 10 times lower; their NRM directions were very close to the present-day geomagnetic field direction in Texas. Since thermal, chemical, and alternating field (af) demagnetization failed to isolate a different magnetization direction, they are excluded from further discussion.

The NRM directions of the samples are shown in Figure 2*a*. The distribution is essentially the same as that published by *Howell and Martinez* [1957], who published NRM results only (not obtained through demagnetization analysis). A pronounced streaking exists between the present-day field direction and directions with ESE declinations and shallow inclinations, suggesting the presence of recent secondary magnetizations. Figure 3 shows the demagnetization diagrams obtained by consecutive af and thermal treatments plotted according to the *Zijderveld* [1967] method. First, af demagnetization up to 1000 Oe was performed, but it failed to eliminate a significant portion of the NRM, as can be seen in all examples of Figure 3. Consequently, thermal treatment was required. Generally, above temperatures of 600°C the trajectory of further demagnetization was linear toward the origin. This last trajectory is interpreted as the removal of a single component of magnetization with a high blocking temperature. This component, in view of its similarity in all samples, is referred to as the characteristic magnetization. The directions of the characteristic magnetizations, after unfolding, are shown in Figure 2*b*.

Generally, the directions of the softer components of magnetization conform to the local present-day geomagnetic field direction (Figures 3*a*–3*c*), and these components are assumed to be of recent origin. Only rarely did this secondary remanence persist above 640°C (Figure 3*c*), but it can be seen that af demagnetization alone is unsuccessful in eliminating it in some samples (Figures 3*a* and 3*c*).

It is of interest to speculate which magnetic minerals carry the secondary and characteristic magnetizations. Potentially magnetic minerals identified were goethite and/or lepidocrocite and hematite. In view of the thermal instability of goethite, lepidocrocite, and (possibly undetected) maghemite above 400°C it is clear that most of the characteristic magnetization and part of the secondary magnetization reside in the fine-

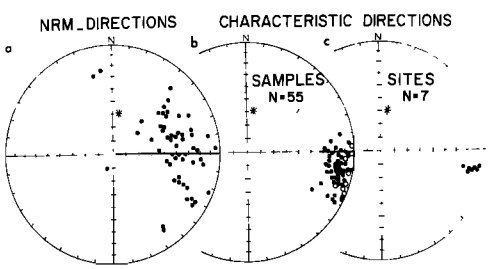


Fig. 2. Equal-area projections of the directions of remanent magnetization of the samples from the Wilberns Formation. Open symbols point upward, and solid symbols point downward; the asterisk is the present-day local geomagnetic field direction. All directions plotted are for samples used in the computation of the final mean direction of magnetization after correction for the minor tilt of the strata.

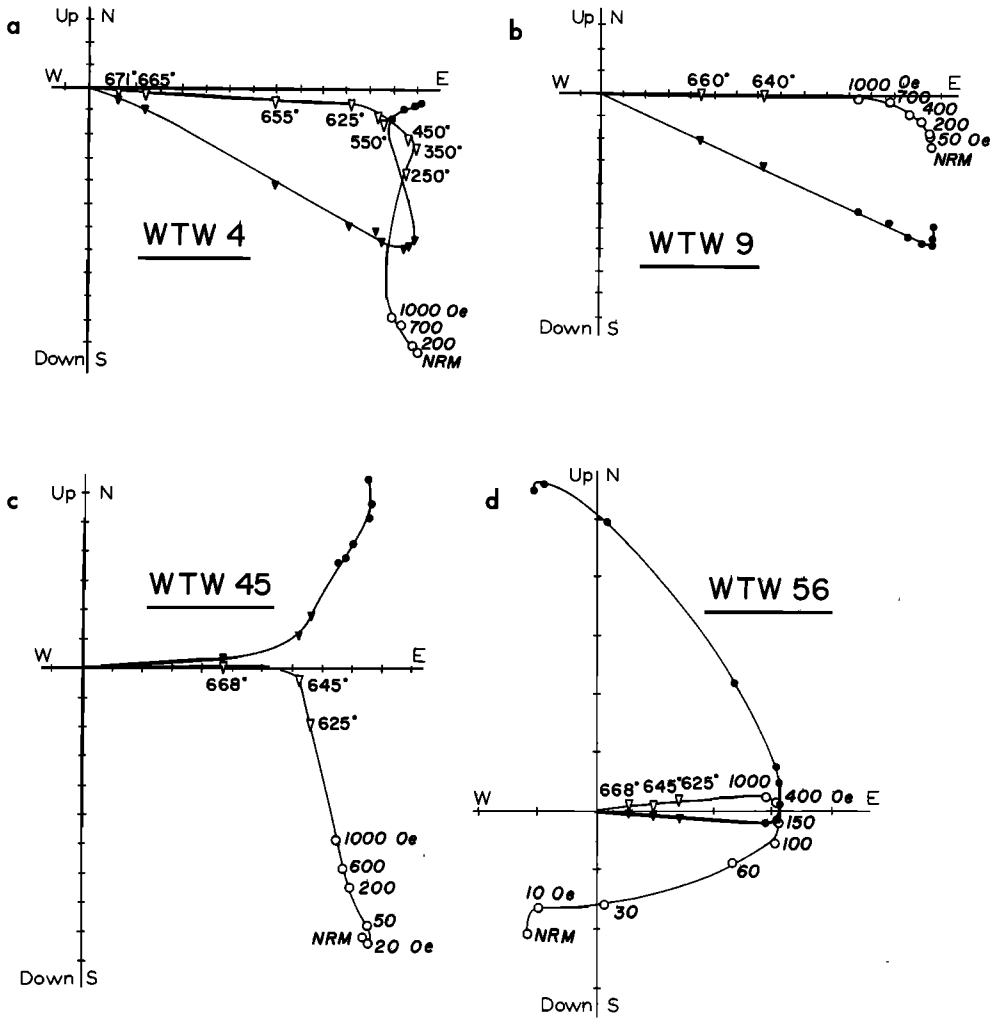


Fig. 3. Orthogonal projections of the successive endpoints of the magnetization vector during progressive af and thermal demagnetization (circles and triangles, respectively). Solid symbols represent projections on the horizontal plane; open symbols those on the east-west vertical plane. Figures 3*a*–3*d* are representative samples from sites 1, 2, 5, and 7, respectively. One unit on either axis corresponds to 1×10^{-5} emu/sample.

grained hematite. It can be assumed that some of the secondary magnetization of the sample of Figure 3*a* is carried by goethite, since it is very resistant to af demagnetization, whereas it is easily eliminated at low temperatures. A chemical demagnetization does not shed much light on the problem, though during a treatment lasting from 12 to 50 hours the specimen of Figure 4 showed elimination of a magnetization similar to that of another specimen of the same core (Figure 3*b*). After 50 hours the calcareous sample crumbled, and the nature of the small portion of remaining magnetization could not be determined.

Results and Comparison With Previous Data

The mean directions of characteristic magnetization for each site are summarized in Table 1 and shown in Figure 2*c*. They cluster very well, with a cone of confidence at the 95% probability level of 4.6°. The mean direction obtained from the seven site means is identical to that of the 55 sample directions (declination D = 101.5°; inclination I = +13.5°) and yields a paleomagnetic pole position at 6.4°S, 338.8°E. There is no reason to suspect that this direction is not of Late Cambrian origin. Assuming this magnetization to be of reversed polarity, we have compared it with the predominantly reversed Late Cambrian directions and pole positions from the Wichita granites, the Lamotte Formation, and the dikes and intrusives from Colorado (Table 2). Several publications describe paleomagnetic and rock magnetic studies of these rocks and suggest a multivectorial nature for their NRM. It cannot be stressed enough that when more than one component of magnetization is present, it is absolutely necessary to determine the directions of the magnetizations that are being removed. Moreover, the direction of the magnetization remaining in a sample after a step in the demagnetization procedure

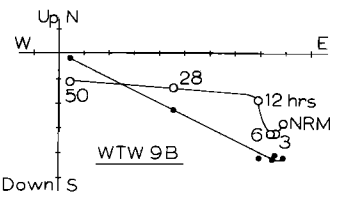


Fig. 4. Orthogonal projection of the successive endpoints of the magnetization vector during progressive chemical demagnetization. Solid symbols represent projections on the horizontal plane, open symbols those on the east-west vertical plane. The specimen is from site 2 and from the same sample as that of Figure 3*b*.

TABLE 1. Summary of the Site Mean Directions of Characteristic Magnetization

Site	R_s	Before Tilt Correction		After Tilt Correction			
		Declination, deg	Inclination, deg	Declination, deg	Inclination, deg	α_{95}, deg	k
1	8/8	101.3	+17.3	102.4	+16.8	12.1	21.8
2	8/8	100.5	+24.8	101.7	+22.4	10.8	27.5
3*	0/7						
4	7/8	100.7	+8.4	100.8	+8.2	11.9	26.8
5	8/8	103.8	+15.5	103.4	+17.1	5.1	117.4
6	8/8	101.5	+11.1	101.5	+11.1	10.9	26.6
7	8/8	99.6	+22.2	99.7	+4.2	10.7	27.8
8	8/8	101.2	+12.6	102.3	+13.3	8.3	45.4
all samples	55/63	...	...	101.7	+13.4	3.6	30.1
all sites	7/8	...	...	101.7	+13.3	4.6	170.9

For the mean pole position from seven site poles, 6.4°N, 158.8°E; α_{95} = 2.6°; and k = 539.1. For the pole position from the mean of seven sites, 6.4°N, 158.8°E; dp = 2.4°; and dm = 4.7°. This polar error (dp, dm) represents the oval of 95% confidence about the pole position. R_s is the ratio of samples (sites) used for computation of the mean of the number of samples (sites) collected, tilt correction is the correction for the dip of the strata, α_{95} is the radius of the circle of 95% confidence, and k is the estimate of the precision parameter [*McElhinny*, 1973, pp. 80–83].

* No characteristic magnetization obtained.

can be meaningless unless it is certain that further decay of the magnetization is that of a single component. Also, spurious magnetizations may sometimes be introduced during af or thermal demagnetization, and this can hamper such an analysis.

The above-mentioned previous studies of Late Cambrian rocks appear to have suffered sometimes from this unfortunate behavior and appear to us of questionable reliability. The Late Cambrian Lamotte Formation was studied by *Al-Khafaji and Vincenz* [1971]. They discovered that the magnetizations showed several distinct groups of directions. A prominent group of their samples did not show appreciable decay of the NRM during af demagnetization up to 1400 Oe (compare with Figures 3*a* and 3*c*), whereas during thermal demagnetization the directional scatter increased for the few samples magnetized strongly enough to be measured at temperatures above 300°C. The Late Cambrian Wichita Mountains igneous complex was studied by *Ku et al.* [1967], *Spall* [1968, 1970], and *Vincenz et al.* [1975]. These collections revealed multivectorial NRM as well and yielded results difficult to interpret. Often the final mean direction for these studies depended on whether af or thermal treatment was used. It is surprising, however, to see the good agreement between most results when the thermally isolated directions of magnetization are used.

Does this immediately imply that af treatment of the Wichita granites failed to yield satisfactory results? The corresponding pole positions (af demagnetization, Table 2) lie on a great circle path toward late Paleozoic pole positions, and it is tempting to invoke partial or complete remagnetization during the late Paleozoic. It is equally possible, however, that we do not yet have complete knowledge about the Cambrian apparent polar wander path for North America. There are indications from Colorado intrusive rocks, though they are imprecisely dated as Cambrian and/or earliest Ordovician, of a pole position near 40°N, 110°E [*Larson and Mutschler*, 1971, Table 2; *French et al.*, 1977]. However, a large portion of the samples studied by French et al. also revealed directions yielding a pole near 12°N, 156°E, close to the Wilberns pole. Furthermore, D. P. Elston (personal communication, 1976) has observed in strata of the Grand Canyon of northern Arizona a 'late Paleozoic' direction in the Muav limestone (Middle Cambrian), which overlies the Bright Angel shale and the Tapeats sandstone (Early and Middle Cambrian). The Tapeats pole position is near that of the Wilberns. The Muav underlies Devonian strata, which also display a late Paleozoic direction. Studies in progress in our laboratory on the Late Cambrian Rome Formation from the Central Appalachians and directions reported for some samples of the Lamotte Formation [*Al-Khafaji and Vincenz*, 1971] appear to lend further support to this trend, but a definite choice between the remagnetization hypothesis and the Cambrian polar wander hypothesis must be deferred until the newer data are in press. Similarly, we postpone the analysis in terms of relative continental drift between North America, Europe, and Gondwanaland for the same reasons.

The preferred pole positions from Table 2, for which full experimental data were available to us, have been plotted in Figure 5. Although some of these results are of questionable reliability, as is explained above, these are the only poles that show mutual consistency so far. Not included are pole positions from Newfoundland and coastal New Brunswick, since these areas conceivably did not form part of cratonic North America during the early Paleozoic [*McKerrow and Ziegler*, 1972].

CONCLUSIONS

A well-clustered group of directions of characteristic magnetization in the Point Peak Member of the Wilberns Formation is interpreted as being representative of the Late Cambrian paleomagnetic field with a reversed polarity. Detailed analysis of thermal, af, and chemical demagnetizations yielded accurate paleomagnetic directions with a dispersion that is within the limits of normal secular variation.

The paleomagnetic pole position of the Wilberns Formation appears to fall very close to reliable Late Hadrynian poles from Canada and is located approximately in the middle of a

TABLE 2. Paleomagnetic Pole Positions

Late Cambrian Poles	N	Declination, deg	Inclination, deg	k	α_{95}, deg	Pole Position
Thermal demagnetization						
Wilberns Formation (600°–670°C, this study)	7	101.5	+13.5	171	4.6	6.4°N, 158.8°E
Lamotte Formation, group 2 only (378°C) [*Al-Khafaji and Vincenz*, 1971]	2	97.5	+17.0	···	···	0.5°N, 168.0°E
Lamotte Formation, groups 1 and 2 (378°C) [*Al-Khafaji and Vincenz*, 1971]	4	114.5	+1.0	6	40.0	18.8°N, 153.4°E
Wichita granites (up to 650°C) [*Vincenz et al.*, 1975]	14	133.5	+10.9	3	25.8	30.5°N, 138.2°E
Wichita granites (500°C) [*Spall*, 1970]	33	97.0	+8.0	3	18.0	4.0°N, 164.0°E
Af demagnetization						
Lamotte Formation, groups 2a and 2b [*Al-Khafaji and Vincenz*, 1971]	42	59.9	+31.9	9	7.7	34.1°S, 183.4°E
Lamotte Formation, groups 1, 2a, and 2b [*Al-Khafaji and Vincenz*, 1971]	3	84.5	+13.2	2	72.3	8.4°S, 177.4°E
Wichita granites (direction and pole recalculated with inclusion of normal polarity sites 4, 10, 11, and 14) [*Ku et al.*, 1967]	12	129.7	+6.7	9	15.3	29.5°N, 141.8°E
Wichita granites [*Spall*, 1970]	33	115.0	+22.0	···	21.0	13.0°N, 147.0°E
Wichita granites [*Vincenz et al.*, 1975]	11	116.2	+17.0	3	34.5	15.8°N, 148.5°E

Preferred Late Hadrynian, Cambrian, and Ordovician Pole Positions for the North American Craton (Plotted in Figure 5)	Pole Position
Beemerville alkaline complex, Late Ordovician [*Proko and Hargraves*, 1973]	35.0°N, 126.0°E
Trenton limestone, Middle Ordovician [*McElhinny and Opdyke*, 1973]	36.0°N, 116.0°E
Colorado intrusives, Cambro-(Ordovician?) (af cleaning) [*Larson and Mutschler*, 1971]	36.6°N, 122.3°E
Colorado intrusives, (Late?) Cambrian (af and thermal demagnetization) [*French et al.*, 1977]	12.0°N, 156.0°E
Wilberns Formation, Late Cambrian (this study)	6.4°N, 158.8°E
Lamotte Formation, Late Cambrian (thermal cleaning, see above)	0.5°N, 168.0°E
Wichita granites, Late Cambrian–525 m.y. (thermal cleaning) [*Spall*, 1970]	4.0°N, 164.0°E
Grenville dikes, Hadrynian [*Murthy*, 1971]	3.0°S, 151.0°E
Coronation sills, Hadrynian–647 m.y. [*Robertson and Baragar*, 1972]	1.0°S, 163.0°E
Franklin diabases, Hadrynian–675 m.y. [*Fahrig et al.*, 1971]	8.0°N, 167.0°E

N is the number of entries used in the statistical analysis (sites or samples), k is the estimate of the precision parameter [*McElhinny*, 1973, pp. 80–83], and α_{95} is the radius of the circle of 95% confidence.

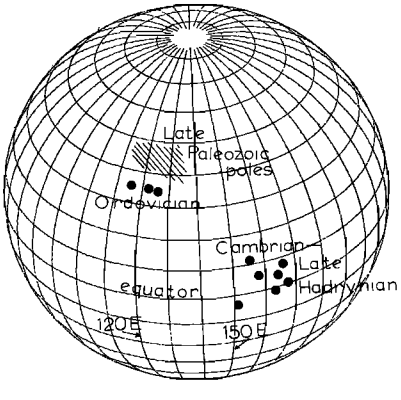


Fig. 5. Preferred virtual geomagnetic pole positions for Late Hadrynian and early Paleozoic times, corresponding to the preferred poles of Table 2.

scattered group of previously published Late Cambrian poles. Inspection of these Late Cambrian results revealed that often their thermally isolated remanent magnetizations showed good agreement with the results of the present study, whereas the directions obtained through af demagnetization appeared to streak more toward the present-day local field directions or toward a middle to late Paleozoic paleomagnetic direction.

The latter trend cannot be unambiguously interpreted at the moment. It appears that the two most likely explanations for this trend are either late Paleozoic remagnetization or a hitherto unexpected Cambrian shift (or loop?) in the apparent polar wander path.

Irrespective of the ultimate choice between these possibilities, we feel that the results from the Wilberns Formation establish a reliable pole position for the Late Cambrian (Middle Croixian). Paleogeographically, this pole implies that the Late Cambrian equator ran from easternmost New Mexico through North Dakota and the western part of Hudson Bay.

Acknowledgments. We thank S. A. Vincenz (St. Louis University), D. P. Elston (USGS, Flagstaff), and several colleagues at the University of Michigan for helpful discussions and critical reading of the manuscript. E. Irving originally suggested this study. This research was supported by NSF grant GA-38038.

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(Received April 12, 1976;
revised May 24, 1976;
accepted June 8, 1976.)